31927 - Applications Development with .NET

32998 - .NET Applications Development

SPRING 2026

ASSIGNMENT-2 COVER SHEET

PROJECT TITLE	Study/Assignment Planner
LEAD STUDENT NAME	Liam Garvey
LEAD STUDENT ID	25051842
GROUP MEMBER NAME(S) AND STUDENT ID(S)	Hayley Pino 14262911 Leon Harmon-Jones 24547248
SUBMISSION DATE	19/10/2026

AI USE RESTRICTIONS FOR PROGRAMMING WORK

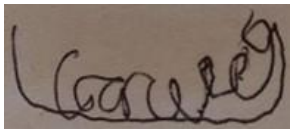
Do not use AI to generate a substantial or complete solution to the programming assignment, or to bypass the learning outcomes of the assessment.

- AI must not be used to invent or misrepresent code execution, test results, screenshots, datasets, performance measurements, external libraries/APIs used, contributions of group members, or any other project detail.
- Permitted support may include concept explanations, syntax help, debugging suggestions, code review, test-case ideas, partial code generation, documentation/comments, and limited code examples where consistent with the assessment instructions.
- You must understand, review, run and test any AI-assisted code yourself, verify that it meets the specification, and be able to explain every submitted component and design decision.
- All AI use must be disclosed. AI output is not an authoritative source and does not replace reading subject materials, documentation, or instructions from teaching staff.

DECLARATION OF ORIGINALITY AND RESPONSIBILITY

X I confirm that the submitted work is the work of the named student(s), except where collaboration, source code, libraries, data, or AI assistance has been appropriately

acknowledged. I understand the submitted program and can explain its implementation, testing and results.

Lead student signature:  Date: 19/10/2026

Group member signature(s): Date: 19/10/2026
L.HJ



AI USE STATEMENT — ASSIGNMENT-2

Attach this completed statement as an appendix to your submission. Disclose all AI-enabled tools used, including uses that did not produce code retained in the final submission.

Student name(s) and ID(s)	Liam Garvey 25051842 Hayley Pino 14262911 Leon Harmon-Jones 24547248
Subject code	X 31927 ☐ 32998
Project / assessment title	ASSIGNMENT-2 Group Project

☐ I did use any AI-enabled tool for this assessment. (If selected, proceed to the declaration.)

1. AI tools and model names

List every AI-enabled tool used (for example ChatGPT, GitHub Copilot, Microsoft Copilot, Gemini, Claude, Grammarly or other coding assistants). Record the model name shown in the tool, or “not shown”.

AI tool or product	Model name (as shown)	Date(s) used

2. What the AI tools were used for (tick all that apply)

☐ Explained programming concepts or APIs	☐ Suggested algorithms, approaches or pseudocode
☐ Helped with syntax or small code examples	☐ Generated or completed code

☐ Debugged errors or explained error messages	☐ Reviewed/refactored code I had written
☐ Suggested unit tests, edge cases or test data	☐ Generated documentation, comments or README text
☐ Helped with data analysis or visualisation code	☐ Assisted with version-control / build / tooling commands
☐ Proofread or revised report text	☐ Other: _____

3. Representative examples of AI use

Give 2–3 examples. Summarise what you asked the tool to do and explain what you accepted, changed, tested or rejected. Do not paste full chat logs unless requested.

Tool / model	Prompt or task summary	How I used, changed or rejected the output	Where it affected the submission

4. Verification and validation

☐ Ran and tested all submitted code and reviewed outputs, errors and edge cases	☐ Checked AI-suggested APIs/libraries against official documentation or course materials
☐ Confirmed that reported outputs, screenshots and performance results came from my/our actual program execution	☐ Reviewed AI-assisted code for correctness, security, originality, licensing/attribution and assessment requirements
☐ I/we can explain every submitted code component and justify the implementation and testing decisions	☐ No confidential, personal or unauthorised third-party material was uploaded to an AI tool

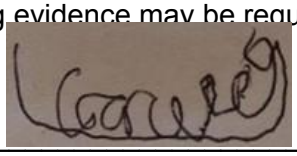
Briefly describe one specific check you performed and any change made as a result:

_____
_____

Declaration

I/we confirm that this statement accurately describes all AI-enabled tools used for this assessment and the extent to which their outputs influenced or appear in the submission. I/we followed the assessment instructions, independently reviewed and tested the submitted work, and take responsibility for the final code, report and results. I/we understand that prompts,

outputs, commit history or other supporting evidence may be requested for an academic integrity review.

Lead student signature:  Date: 19/10/2026

Group member signature(s): ____L.HJ_____ Date: 19/10/2026

Hayley Cira

University Study / Assignment Planner

Project Summary

Project Motivation

As a group of students, having a study tool that also allows us to view details about our classes and assignments would be very helpful, that we felt would be both relevant to us as individuals, and helpful to a decent sized amount of the population (i.e high school and university students). Normal study planner tools don't generally have information on school subjects, and the uni timetable only shows general information about classes, so we thought the ability to combine the two features into one product would be a helpful tool. We wanted to create something that would be helpful to students, both during high stress periods when lots of assignments are due, and in lower stress periods when planning for upcoming assignments/study periods.

Purpose

The idea of this project is to create a tool that can assist students when managing their time, planning for assignments, and helping them to keep track of how they are progressing in relation to individual assignments. The implementation of custom milestones allows them to define how they want to approach each assignment, giving them the chance to choose when and how they want to progress towards completion. The study planner section also gives them the ability to plan out study sessions in the style that works best for them, rather than regimented structures. It also allows them to easily keep track of both past and upcoming study sessions, allowing them to both track how much progress they've made, what they still have to work on, and what direction they want to take next

Key Features

- Students can view all of their subject details, including lectures, classes such as tutorials and workshops and details of those classes, such as the time they are on, their location, and any notes the student wishes to add
- Students can also view their assignments and the details of those assignments such as its type, due dates and any notes about the assignment. Students can also create custom milestones that allow them to define how they want to approach each assignment, and track their progress based on the standards that they set themselves. Assignments can be viewed from their respective subject pages, and from an assignment pager showing a list of all assignments.
- Students can also add assignments whenever they want, and can modify those assignments the same way they can current assignments
- Students can also create and delete study sessions, with the ability to view details about past and future study sessions

Development Approach

Windows form was used to construct the project, and to visually represent the databases containing the subject, assignment and study session information, as well as the code programs to manipulate and sort that data.

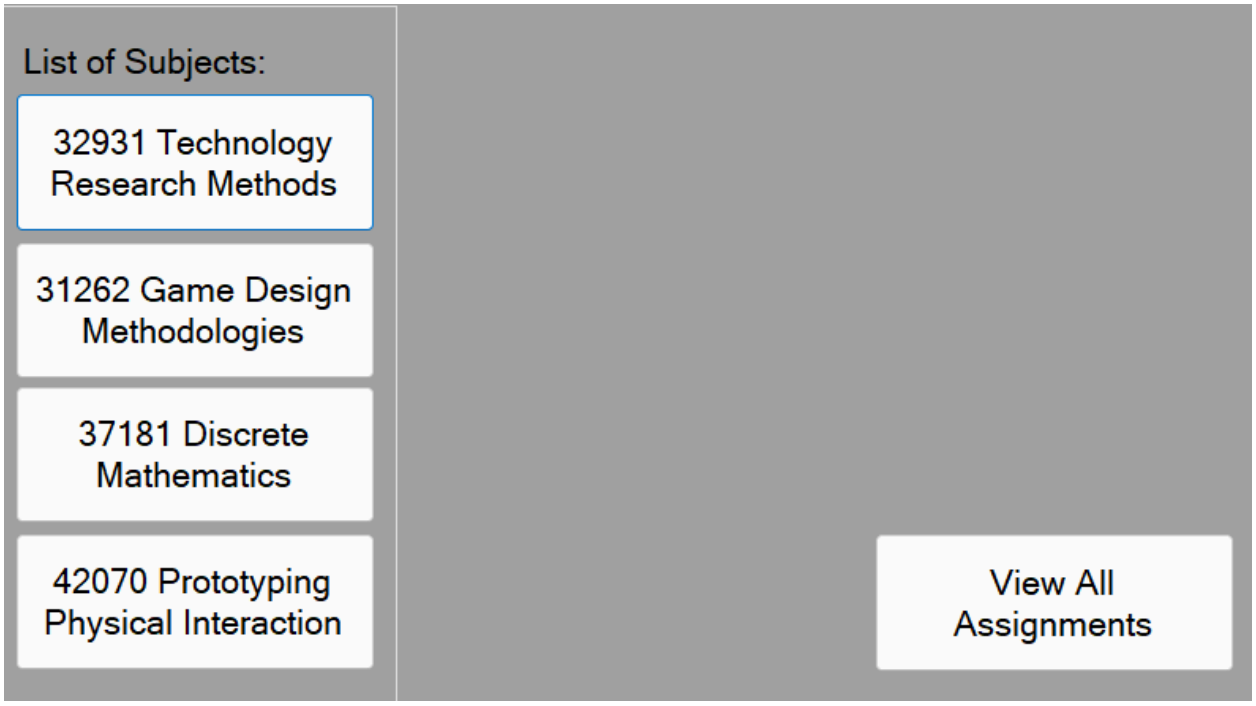
For the subject pages and the main page, rather than using different forms, each subject was assigned a corresponding panel. The home page was also assigned a panel. Each panel, and all of the different components contained within the panel that represented the individual

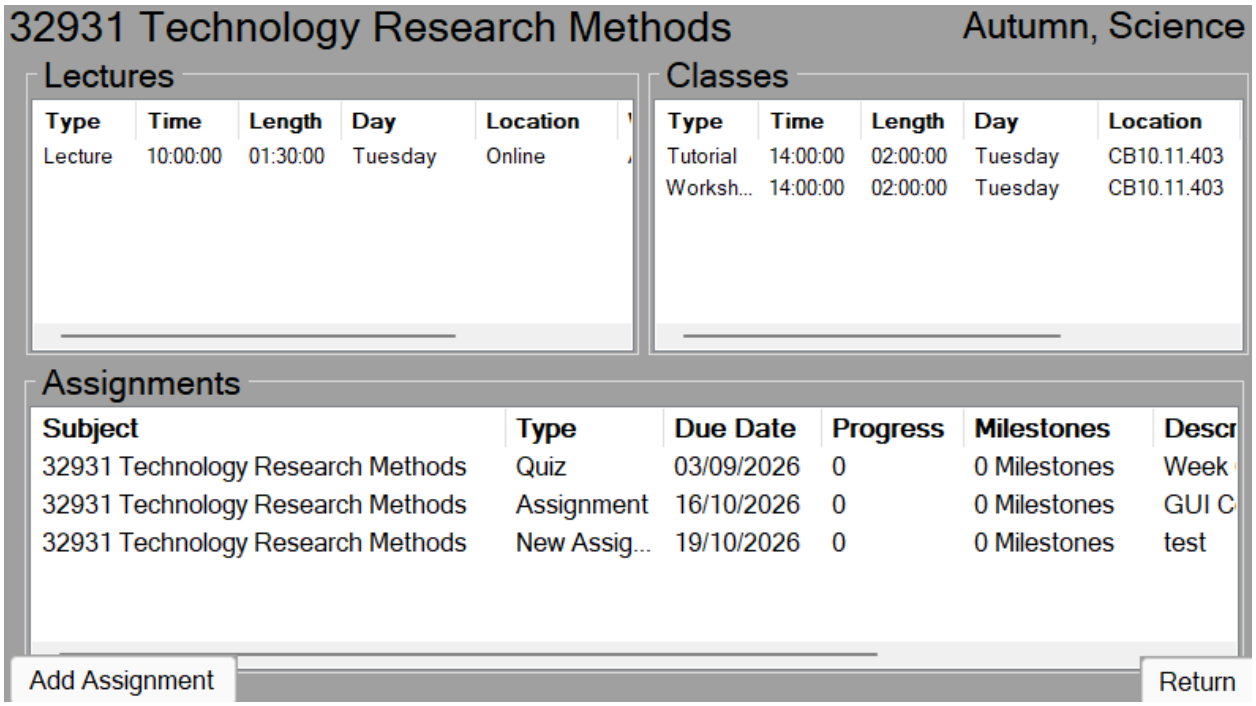
subjects, started off as invisible, are turned visible when their associated button is selected, then turned invisible again when the user navigates away from the panel.
To fill out all of the different subject and assessment information, a csv file filled with dummy information is created. Then on project initialization, all of the subject classes, and their associated classes, lectures, assignment classes are created by reading from the csv file and sorting the data it contains into the relevant data fields of said classes. The different subject panels are then grabbed during the initialization of the form, and the group boxes which display the different subject data such as classes, lectures and assignments are grabbed, then iterated through to display a summary of all of the different parts of the subject. All of the assignment information is placed onto buttons, which open a different form representing all of the assignment data, and allow users to create, modify, and remove custom milestones to the assignment.
After the user closes the program, the updated subjects, including all of their classes, and assignments are converted into a csv file and used to overwrite the original csv file used at the start of the program. This is done to save any changes that the user makes to the classes while using the program.

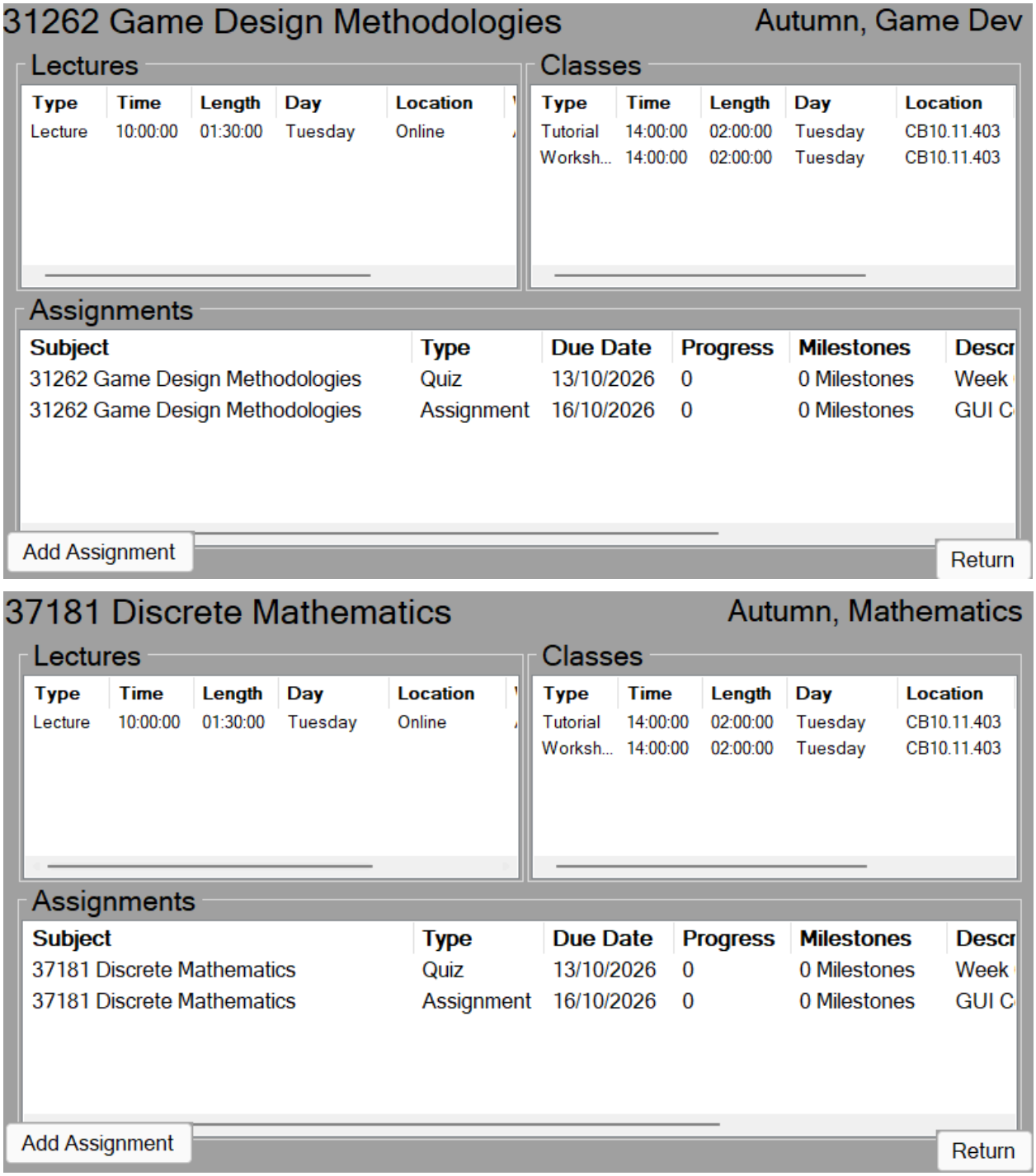
The assignment details form is reused for both editing existing assignments and creating new assignments. A save callback determines whether the completed assignment is added to the selected subject or applied to an existing assignment.
When an assignment is opened, the form creates a temporary copy of the assignment and its milestones. This allows users to discard edits without changing the original data. Validation checks prevent blank assignment types and milestone names, while unsaved change prompts allow users to save, discard or continue editing before closing the form.

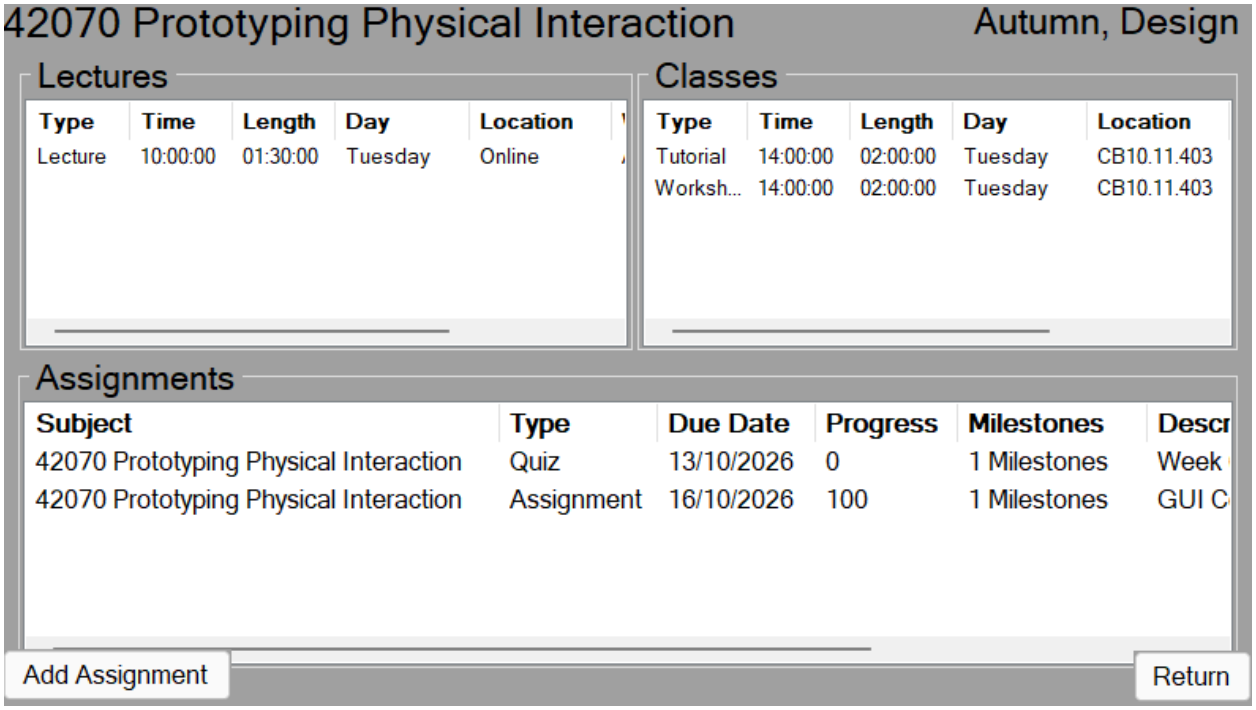
For the study session feature, it was designed mostly self-contained with its own separate window rather than a panel to distinguish itself from the subject and assignment classes. In regards to the UI , each section of the Study Session window is placed in a group box with UI elements being docked into the parent container. There are three main sections of the Study Session including the “Add Session”, “Study Log” and “Upcoming Session”. The Study Session class consisted of members which made up a study session object and specific formatting for strings consisting of dates and time. These session objects are made into a list in the Study Session form. In order for the user to add a study session, they must input all the values otherwise they would receive a message box warning of what missing or invalid input they have. The values are made so that users are only able to create a study session that consists of a future time, date and a study duration greater than 0. Once a study session is successfully made after confirmation with the user, the session is added to the “Upcoming Session” section. With the listed details on the ListView, users are able to select a session and once selected pick the green “Session Completed” button or the red “Remove Session” button. By pressing the “Session Completed”, after confirming with the user the session is moved into the Study Log which can only be viewed. The same applies to the “Remove Session” button. Each list view are filtered using LINQ.

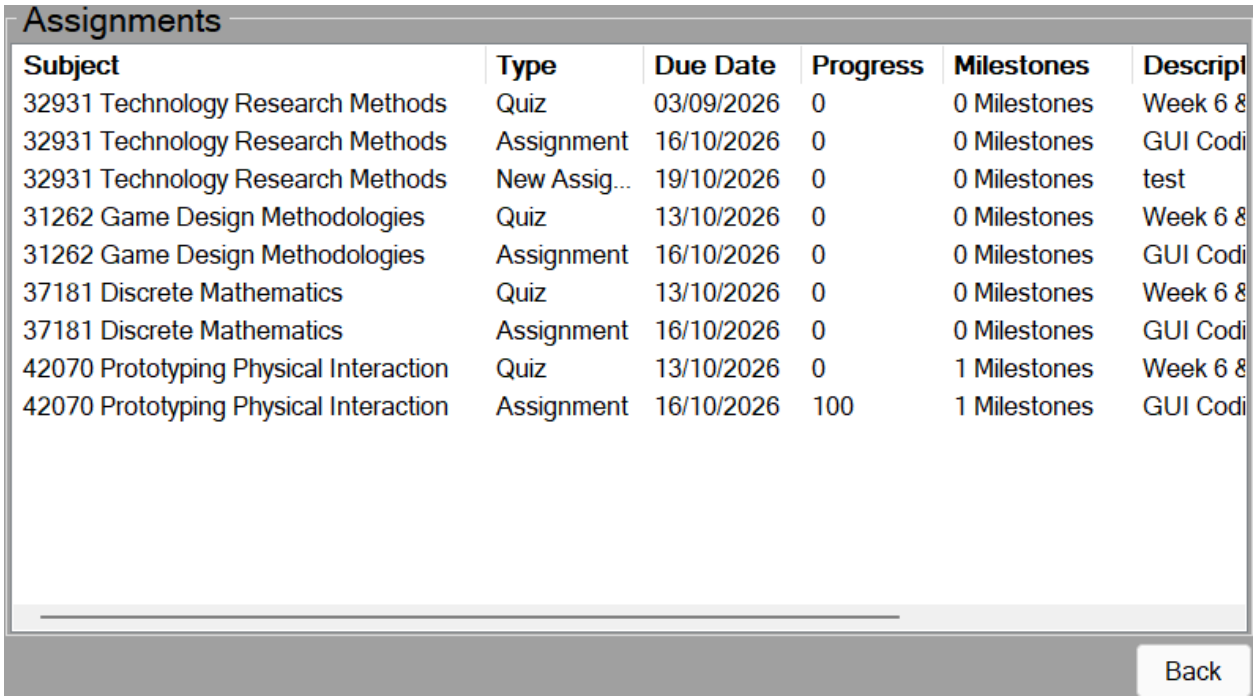
Interface Design

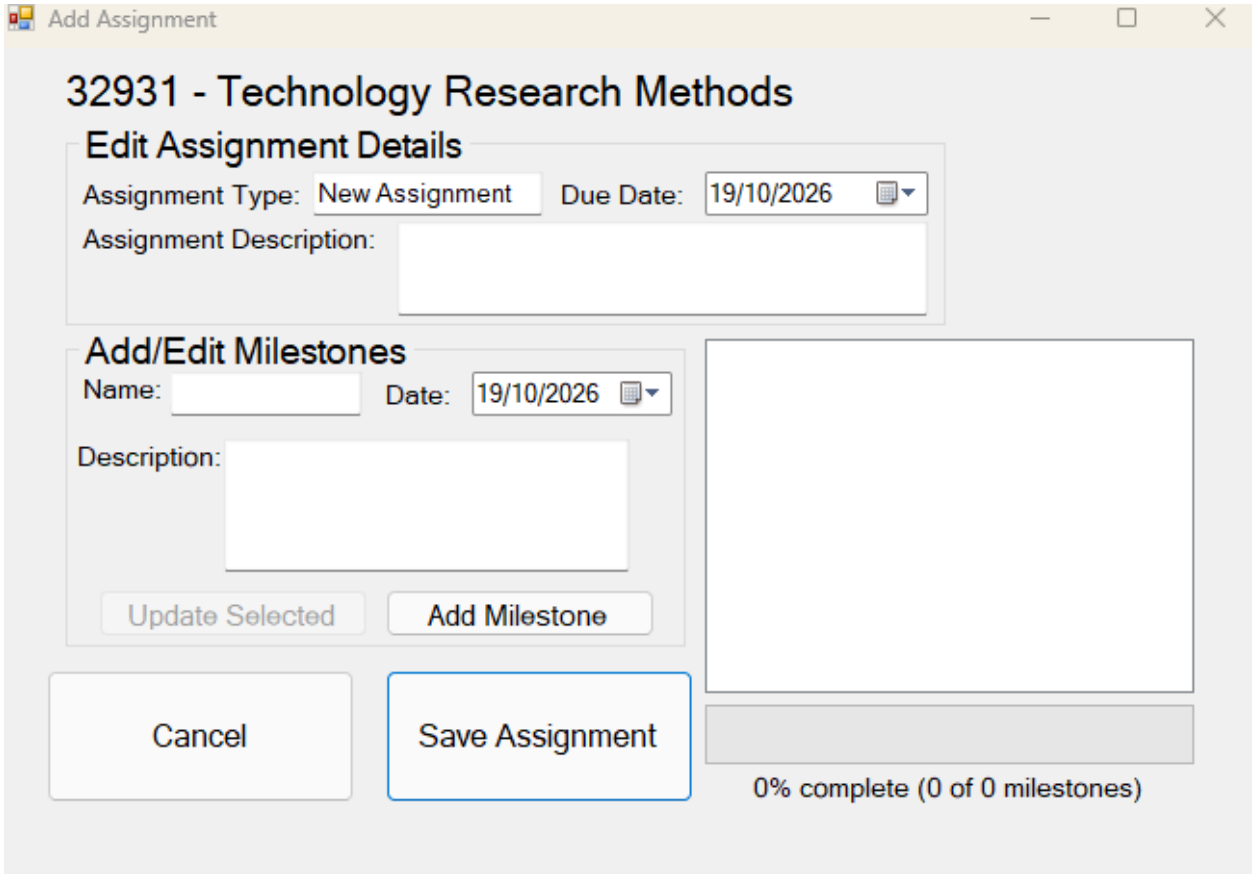


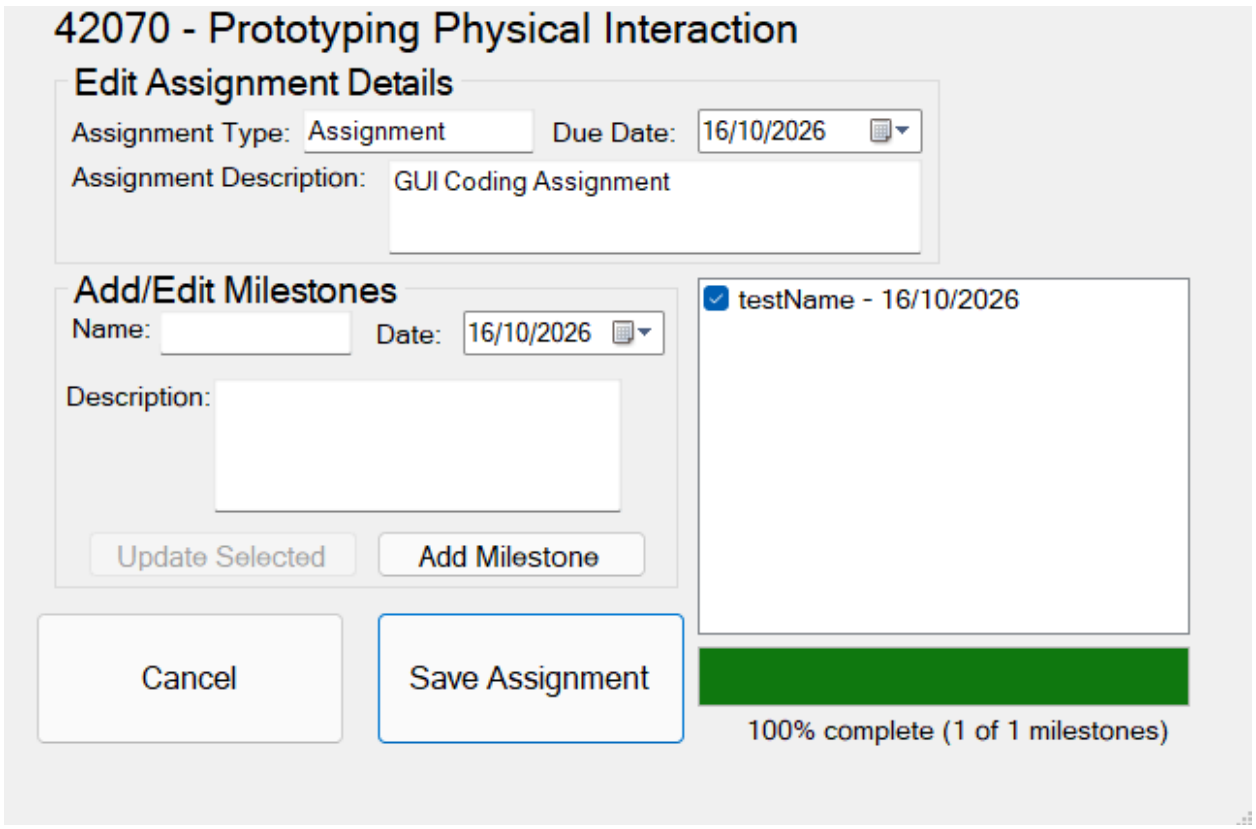


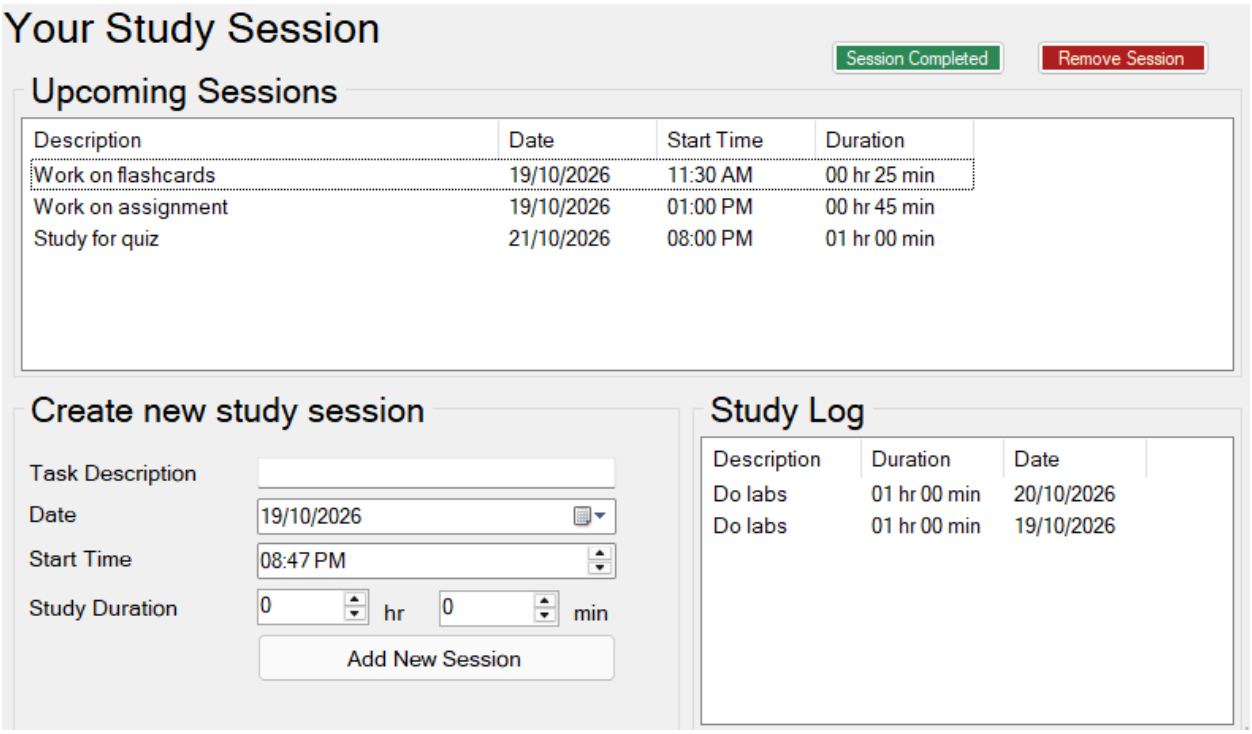

31262 Game Design Methodologies
Autumn, Game Dev
Lectures
Type Time Length Day Location
Lecture 10:00:00 01:30:00 Tuesday Online
Classes
Type Time Length Day Location
Tutorial 14:00:00 02:00:00 Tuesday CB10.11.403
Worksh... 14:00:00 02:00:00 Tuesday CB10.11.403
Assignments
Subject Type Due Date Progress Milestones Descr
31262 Game Design Methodologies Quiz 13/10/2026 0 0 Milestones Week
31262 Game Design Methodologies Assignment 16/10/2026 0 0 Milestones GUI C
Add Assignment
Return
37181 Discrete Mathematics
Autumn, Mathematics
Lectures
Type Time Length Day Location
Lecture 10:00:00 01:30:00 Tuesday Online
Classes
Type Time Length Day Location
Tutorial 14:00:00 02:00:00 Tuesday CB10.11.403
Worksh... 14:00:00 02:00:00 Tuesday CB10.11.403
Assignments
Subject Type Due Date Progress Milestones Descr
37181 Discrete Mathematics Quiz 13/10/2026 0 0 Milestones Week
37181 Discrete Mathematics Assignment 16/10/2026 0 0 Milestones GUI C
Add Assignment
Return


42070 Prototyping Physical Interaction
Autumn, Design
Lectures
Type Time Length Day Location
Lecture 10:00:00 01:30:00 Tuesday Online
Classes
Type Time Length Day Location
Tutorial 14:00:00 02:00:00 Tuesday CB10.11.403
Worksh... 14:00:00 02:00:00 Tuesday CB10.11.403
Assignments
Subject Type Due Date Progress Milestones Descr
42070 Prototyping Physical Interaction Quiz 13/10/2026 0 1 Milestones Week
42070 Prototyping Physical Interaction Assignment 16/10/2026 100 1 Milestones GUI C
Add Assignment
Return

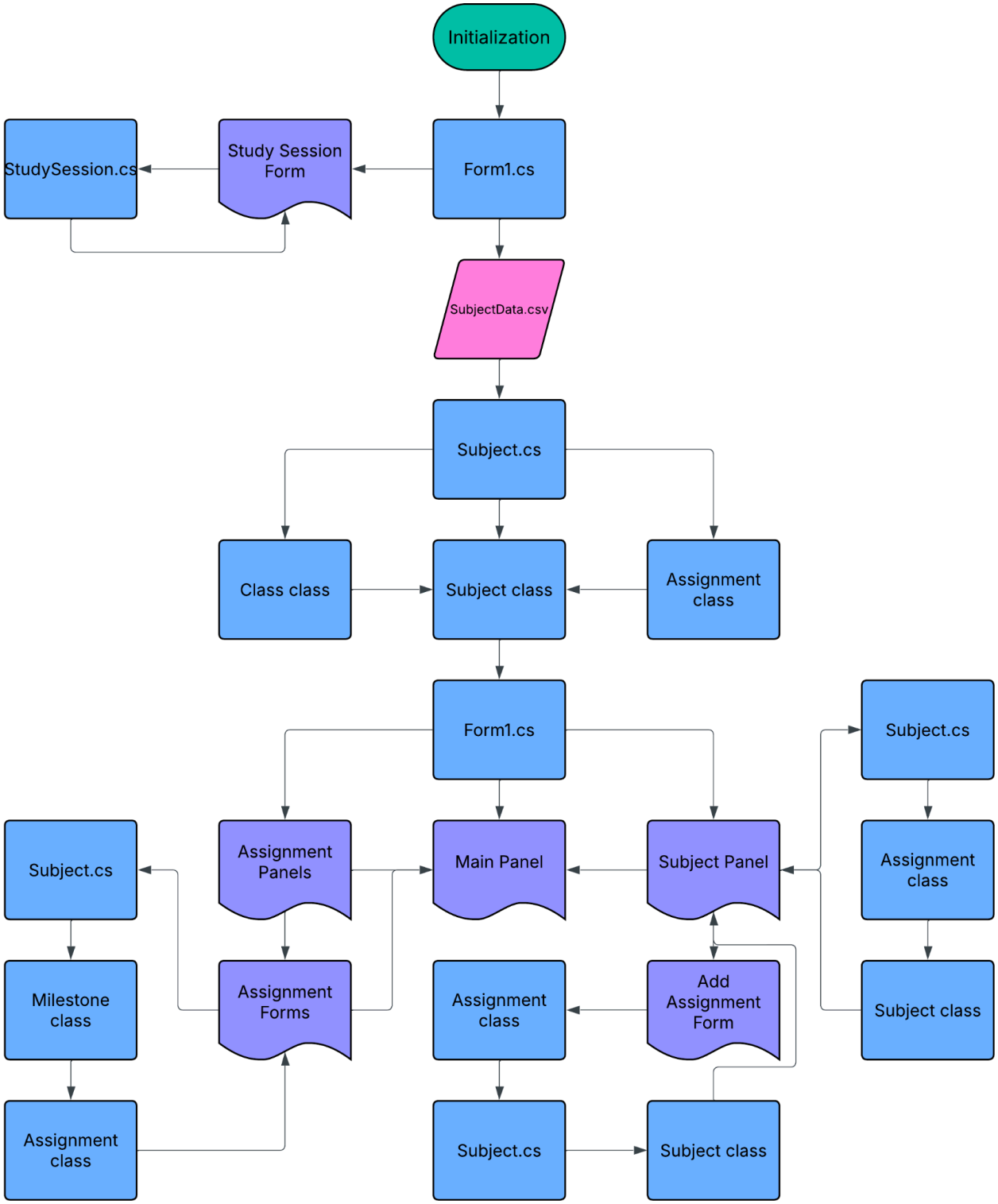

Assignments
Subject Type Due Date Progress Milestones Descripl
32931 Technology Research Methods Quiz 03/09/2026 0 0 Milestones Week 6 &
32931 Technology Research Methods Assignment 16/10/2026 0 0 Milestones GUI Codi
32931 Technology Research Methods New Assig... 19/10/2026 0 0 Milestones test
31262 Game Design Methodologies Quiz 13/10/2026 0 0 Milestones Week 6 &
31262 Game Design Methodologies Assignment 16/10/2026 0 0 Milestones GUI Codi
37181 Discrete Mathematics Quiz 13/10/2026 0 0 Milestones Week 6 &
37181 Discrete Mathematics Assignment 16/10/2026 0 0 Milestones GUI Codi
42070 Prototyping Physical Interaction Quiz 13/10/2026 0 1 Milestones Week 6 &
42070 Prototyping Physical Interaction Assignment 16/10/2026 100 1 Milestones GUI Codi
Back


Add Assignment
32931 - Technology Research Methods
Edit Assignment Details
Assignment Type: New Assignment
Due Date: 19/10/2026
Assignment Description:
Add/Edit Milestones
Name:
Date: 19/10/2026
Description:
Update Selected
Add Milestone
Cancel
Save Assignment
0% complete (0 of 0 milestones)


42070 - Prototyping Physical Interaction
Edit Assignment Details
Assignment Type: Assignment
Due Date: 16/10/2026
Assignment Description: GUI Coding Assignment
Add/Edit Milestones
Name:
Date: 16/10/2026
Description:
Update Selected
Add Milestone
testName - 16/10/2026
Cancel
Save Assignment
100% complete (1 of 1 milestones)


Your Study Session
Session Completed
Remove Session
Upcoming Sessions
Description
Date
Start Time
Duration
Work on flashcards
19/10/2026
11:30 AM
00 hr 25 min
Work on assignment
19/10/2026
01:00 PM
00 hr 45 min
Study for quiz
21/10/2026
08:00 PM
01 hr 00 min
Create new study session
Task Description
Date
19/10/2026
Start Time
08:47 PM
Study Duration
0
hr
0
min
Add New Session
Study Log
Description
Duration
Date
Do labs
01 hr 00 min
20/10/2026
Do labs
01 hr 00 min
19/10/2026

Flowchart



How to Use

Launching the program displays the home screen. The home screen contains lists of enrolled subjects, planned study sessions and a link to a page displaying all assignments. Selecting a subject opens a page which displays all information about the subject, including its classes,

lectures and assignments. Selecting an assignment, either on the subject panel or the view all assignments panel, opens a page displaying all information about the assignments. While on a subject page, users can add new assignments. While on an assignment page, users can edit the assignment details, and create, edit/modify and remove milestones from the assignment. While on a subject page, users can create a custom assignment, edit as they want, then add it to the corresponding subject page. Users can access the study session page from the home screen. Using that page, they can view past and upcoming study sessions, create and remove study sessions, and mark study sessions as completed

Team Members Contribution

Liam Garvey

My main role in the group was writing the report and creating the Subject portion of the application. I created all of the subject pages, and I also assisted in creating the initial assignment and study session classes, and gave the application the ability to both read and store data related to the subjects, assignments and milestone data being manipulated by users of the application.

- Wrote the Project Motivation, Purpose, Key Features, Project Summary, How to Use, Flowchart, Interface Design sections and part of the Development Approach segment of the report.
- Created all aspects of the Subject and Class classes
- Created the initial version of the Assignment and Milestone classes, including their member variables and class constructors
- Created and edited the AssignmentInfo and AddMilestone methods of the assignment class
- Updated the constructor methods and the SetComplete method of the Milestone class
- Created the SubjectData.csv file to load the program with data
- Created the LoadSubjectData and SaveSubjectData methods read from a csv file to create a list of subjects, then transform that list back into a csv function after application close to ensure data persistence
- Created all four Subject panels, View Assignments panel, and the Home panel, and all corresponding functionality, UI elements and code elements for those panels
- Created the DisplaySubjectData, FillGroupBox, FillAssignmentList, ChangePanel, LoadAssignment, AddAssignment methods, all button click methods, and all index changed methods in the Form1.cs file
- Wrote the ReadMe File
- Created the two Interfaces, Generic Method, and Generic Collections

Hayley Pino

- Main role in the group was mostly handling the UI design and handling the study session feature of our application. For our application
- Designed the initial user flow of the application based on the project proposal.
- Roughly design and map out the required interfaces and components needed for our application.

- Our main four forms, I proposed were the main menu window, study session window, subject window and assignment window. The UI design of each window was completed by division of work. The UI of each window was then designed depending on the division of work (ie. I was in charge of the Study Session feature so I designed the UI of the Study Session window).

Figure 1: Initial Interface Design and User Flow (Refer to References)

- The main requirements of the study session feature was for users to be able to plan an upcoming study session which can be put into a list and when completed can be checked off into a study log list.
- I also finalised the UI so that all the components in each interface would be reactive to resizing the window.
- This entire study session feature was created and designed by Hayley.

Leon Harmon-Jones

My main role in the group was developing the assignment and milestone features. I worked on the Assignment class and Assignment Details Form, including the interface and code used to manage assignment details. The form displays the subject the assignment belongs to and allows users to edit its type, due date and description, as well as add/edit milestones.

For milestones, I implemented functionality for users to add milestones, edit their names, descriptions and expected dates, as well as mark them as complete when they have finished a milestone. Assignment progress is calculated from the proportion of completed milestones for that assignment. I also added validation so users receive a message when required information, such as an assignment type or milestone name is missing.

I implemented the Save and Cancel behaviour using a temporary copy of the assignment and its milestones. This allows users to make changes without immediately changing the original assignment. Saving applies those changes, while discarding leaves the original untouched. The form also prompts users when they try to close with unsaved changes.

I connected these features to the assignment views so that both the subject's assignment list and the all assignments list update after changes are saved. I also implemented the functionality behind the Add Assignment buttons on each subject page, allowing users to create an assignment for the currently selected subject page reusing the same assignment details form.

Figures

Figure 1.Initial Interface Design and User Flow

